APPROXIMATION DESCENDS FROM AN OPEN SUBSCHEME CONTAINING THE SUPPORT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For an open immersion j the extension by zero j_! agrees with Rj_* and is compatible with restriction to opens. invoked at proof lines 3876-3881. Identifies the pushforward of the approximation with j_!(E|_U).
- The pushforward of a perfect complex along an open immersion is perfect. invoked at proof lines 3880-3881. Keeps the approximation perfect over X.

Source reference. Original definition 08EM: sources/stacks-perfect/pinned-original-source.tex, source lines 3814–3814. Cited at statement lines 3864-3865 to define the triples; the descent argument is inside the counted span.

Source reference. Original lemma 08DF: sources/stacks-chow/tags.txt, source lines 10635–10635. Cited at proof lines 3880-3880 to identify j_!(E|_U); the conclusion is inside the counted span.

Source reference. Original lemma 08DP: sources/stacks-chow/tags.txt, source lines 10644–10644. Cited at proof lines 3880-3881 to keep the approximation perfect; the conclusion is inside the counted span.

Lemma 0.2. *Let X be a scheme. Let $U \subset X$ be an open subscheme. Let (T, E, m) be a triple as in Definition 08EM. If*

(1) *$T \subset U$,*
(2) *approximation holds for $(T, E|_U, m)$, and*
(3) *the sheaves $H^i(E)$ for $i \geq m$ are supported on T,*

then approximation holds for (T, E, m).

Proof. Let $j : U \to X$ be the inclusion morphism. If $P \to E|_U$ is an approximation of the triple $(T, E|_U, m)$ over U, then $j_!P = Rj_*P \to j_!(E|_U) \to E$ is an approximation of (T, E, m) over X. See Cohomology, Lemmas 08DF and 08DP. □

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